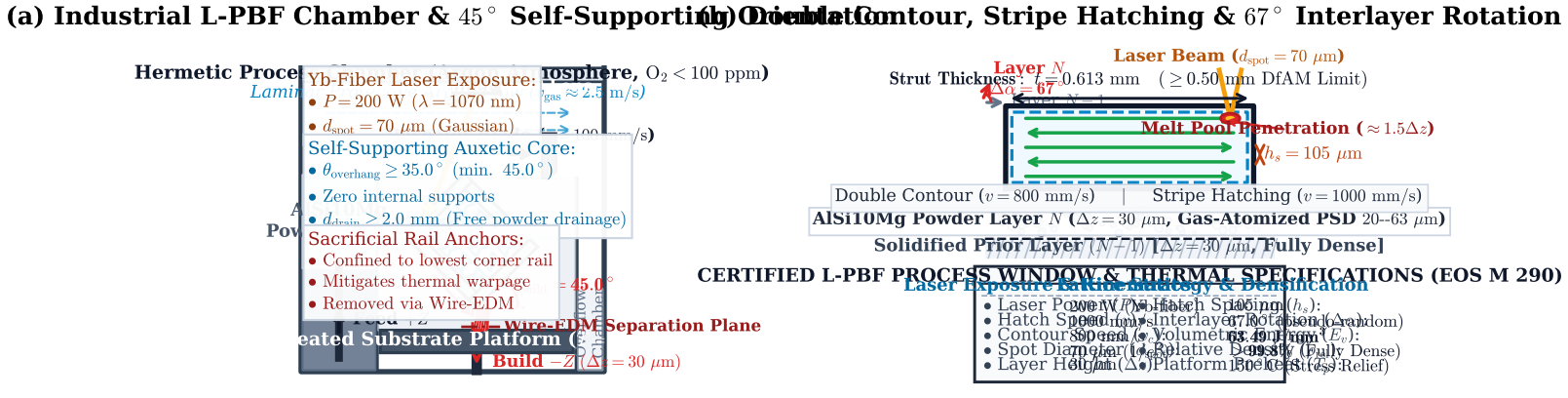

(a) Industrial L-PBF Chamber & 45° Self-Supporting Double Contour, Stripe Hatching & 67° Interlayer Rotation
Hermetic Proce
tosphere, O2 < 100 ppm)
Yb-Fiber Laser Exposure:
P = 200 W (λ = 1070 nm)
dspot = 70 μm (Gaussian)
Self-Supporting Auxetic Core:
θoverhang ≥ 35.0° (min. 45.0°)
Zero internal supports
ddrain ≥ 2.0 mm (Free powder drainage)
Sacrificial Rail Anchors:
Confined to lowest corner rail
Mitigates thermal warpage
Removed via Wire-EDM
= 45.0°
Wire-EDM Separation Plane
eated Substrate Platform (
Build −Z (Δz = 30 μm)
Laser Beam (dspot = 70 μm)
Layer N
Δα = 67°
Strut Thickness: t = 0.613 mm (≥ 0.50 mm DfAM Limit)
Melt Pool Penetration (≈ 1.5Δz)
hs = 105 μm
Double Contour (v = 800 mm/s) | Stripe Hatching (v = 1000 mm/s)
AlSi10Mg Powder Layer N (Δz = 30 μm, Gas-Atomized PSD 20–63 μm)
Solidified Prior Layer [N−1] (Δz = 30 μm, Fully Dense)
CERTIFIED L-PBF PROCESS WINDOW & THERMAL SPECIFICATIONS (EOS M 290)
Laser Exposure Baking Strategy & Densification
Laser Power (P): 200 W
Hatch Spacing (hs): 105 μm
Contour Speed (v): 800 mm/s
Spot Diameter (dspot): 70 μm
Layer Height (Δz): 30 μm
Interlayer Rotation (Δα): 67° (Golden Angle)
Volumetric Energy (Ev): 63.49 J/mm³
Relative Density: 99.8% (Fully Dense)
Platform Heating: 150 °C (Stress Relief)